

MP#1 - Connected Component Labeling Report

Algorithm Implementation

As discussed in class, the code performs connected component labeling using a two-pass sequential approach, with 4 connected pixels/blocks, checking the pixels directly above and to the left of the current coordinate. During the first pass, the script assigns preliminary labels. When the scanner finds a foreground pixel with no labeled neighbors, it assigns a new integer ID and records it as a new set in the 'eqs' list.

If the pixel has neighbors that already have labels, it adopts the minimum label value. If these neighbors have different labels, the code identifies the sets containing those values within the 'eqs' list. It then uses the 'update' method to merge the sets and the 'remove' method to delete the redundant group. This process tracks label equivalencies without using complex tree structures. The second pass uses a dictionary created from the final state of these sets to map all temporary labels to sequential integers.

Image Preprocessing

The implementation uses the Pillow library. Each bitmap is loaded and converted to an 8-bit grayscale format. The script applies a threshold of 128 to convert these grayscale values into a binary matrix, as it is the midpoint of the 0 to 255 intensity scale, hence providing a reliable cutoff for the given test images. This conversion creates the 0 and 1 integer binary data necessary for the labeling logic to distinguish objects from the background.

Results and Filtering

Testing on "test.bmp" and "face.bmp" produced distinct labels for all visible shapes. For "gun.bmp", a size-based filter was applied to manage noise components. Initial trials with a threshold of 50 pixels caused two small objects to disappear from the result. Lowering this threshold to 25 pixels allowed those components to remain visible while still providing a mechanism for smoothing. The script exports the final matrices to CSV files to verify the integer assignments, in a manner we saw in class. The visualization plots use the "gist_rainbow" map and treat zeros as null values to keep the background white.